DoodleSim

Terrain-Aware RF Planning Suite for Doodle Labs Mesh Rider Networks

Technical Summary • July 29, 2026 • Internal — Doodle Labs Sales Engineering

1. Executive Overview

DoodleSim is a browser-based, GIS-style planning suite for Doodle Labs Mesh Rider deployments. A user places radios on a map, configures each install (radio model, band, antenna, height, cabling, amplification), and receives terrain-aware link-budget analysis: line-of-sight and Fresnel-zone checks over real elevation profiles, per-MCS throughput predictions, coverage heatmaps, goal-driven equipment recommendations, and customer-ready exported design documents.

The tool is live at **doodlesim.jus2419497.workers.dev** (Cloudflare Workers, free tier), with source at **github.com/jus24194971/DoodleSim**. It is currently in internal review prior to any customer-facing rollout. Its link-budget engine reproduces the official Doodle Labs Range Estimation Tool exactly in free-space mode, then extends it with terrain physics the official tool does not model.

2. System Architecture

Layer	Technology	Notes
Frontend	Vite + vanilla ES modules + MapLibre GL JS	Single-page app; OSM / Esri satellite basemaps + terrain hillshade
Terrain data	AWS Terrarium elevation tiles (SRTM-derived, ~30 m)	Free, no API key; decoded client-side from PNG tiles; cached per tile
RF engine	Pure JavaScript (src/engine.js, coverage.js, advisor.js)	All computation client-side; coverage rays in <1 s
Backend	Cloudflare Worker + D1 (SQLite)	Community antenna library API (/api/antennas); static assets served by the same Worker
Persistence	localStorage autosave + portable .json layout files	No server-side storage of user layouts
Hosting / deploy	Cloudflare Workers, wrangler CLI	Static build + Worker deploy in one step; $0/month at current usage

3. Data Foundation

3.1 Radio catalog

Fourteen radio variants (nanoOEM/miniOEM/OEM/Wearable v3 and v4, RM-5200 both sub-bands, RM-1300 and RM-1400 quad-band pairs) with per-MCS transmit power arrays and the shared sensitivity curve (-87 to -69 dBm, MCS0-7 at 20 MHz), extracted directly from the official Range Estimation Tool with the behavioral rules verified by black-box probing: combined TX = min(configured, per-chain + 3 dB), MCS8-15 pay a +3 dB sensitivity penalty, and sensitivity scales 10*log10(BW/20). Public datasheet data for 15 models (with full MCS tables) is retained separately; the estimator numbers are ~6 dB more conservative and are used as the sales-safe default.

3.2 Antenna catalogs (three tiers)

Tier	Count	Provenance
Verified third-party	53 usable (from 90 sheet rows)	Every row of the source spreadsheet checked against manufacturer datasheets by parallel research agents; 31 confirmed, 19 corrected, 20 exposed as non-products, 6 unverifiable
Doodle Labs recommended (star)	38	Official KB Recommended Antenna Matrix, incl. Doodle ANT-915/1700/2025/2450-3-O multipolarized omnis
Community (cloud)	user-submitted	Validated submissions via the shared library API; deduplicated; plausibility-checked

Notable verification findings: most L-com HG4958-xx part numbers in the source sheet do not exist as written; both Poynting entries were cellular (not Wi-Fi) antennas; the Triad TTRM2005D BDA is rated 25 W, not 16 W; and one recommended sector antenna (5 W max input) is incompatible with the 20 W BDA it was paired with.

3.3 Supporting equipment models

Coax model: loss(dB/100 m) = k*sqrt(f_MHz) fitted to published LMR-class tables (195/240/400/600 classes, i.e. MC400/MC600 equivalents), plus 0.5 dB connector allowance — computed per node from cable type, length, and operating frequency. BDA support covers the Doodle Labs Boost BDA (4400-6400 MHz, 4 W, +13 dB, 2x2) and generic Triad-class amplification. Custom antennas support seven type templates with physics-based dish estimation: gain = 10*log10(0.55*(pi*D/lambda)^2), HPBW = 70*lambda/D.

4. RF Modeling Methodology

4.1 Link budget

RSSI = TX(MCS) + G_tx + G_rx - L_cable - L_pattern + G_bda - FSPL - L_diffraction. A link closes at a given MCS when margin >= 10 dB fade margin (both directions evaluated; the worse governs). FSPL(dB) = 32.45 + 20*log10(d_km) + 20*log10(f_MHz).

4.2 Terrain analysis

Each link samples a 160-point elevation profile. Earth curvature uses the 4/3-effective-radius model. Clearance is tested against 60% of the first Fresnel zone; the worst obstruction feeds the ITU-R P.526 single knife-edge approximation: J(v) = 6.9 + 20*log10(sqrt((v-0.1)^2+1) + v - 0.1) for v > -0.78. Coverage heatmaps ray-cast 180 azimuths x 120 steps with cumulative worst-knife-edge per ray.

4.3 Antenna patterns

Parametric 3GPP-style pattern in both planes: L = min(12*(theta/HPBW)^2, floor), floors 25 dB azimuth (front-to-back) and 20 dB elevation, combined and capped at 30 dB. Off-axis angles derive from link geometry: bearing for azimuth, and elevation angle from terrain heights for the vertical cut. Catalog HPBW values were captured during datasheet verification.

4.4 Near-ground empirical model

The official estimator's "ground level" mode was probed extensively: distance slope is exactly 22.25 dB/decade at fixed frequency, with a frequency intercept A(f) sampled at eight frequencies (250-5800 MHz) and interpolated in log-f. Used by the Plan Advisor for UGV/handheld/vehicle scenarios, cross-checked against the KB's official UGV worked examples. The exact closed form remains one open item (code extraction was blocked by browser DLP); the empirical fit tracks official outputs within ~3-4%.

5. Validation

Test vector	Official result	DoodleSim	Status
miniOEM v4, 2450 MHz, 20 MHz, 3+3 dBi, MCS8	3879.2 m	3879.2 m	exact
Bandwidth scaling 10/20/40 MHz (all MCS)	sqrt(2) range steps	identical to the meter	exact
Power clamp / attenuation / BDA arithmetic	probed matrix	identical to the meter	exact
KB UAV ex.: 2450 MHz, 10 MHz, 3+3 dBi @ 5 Mbps	4357.7 m	4357.7 m	exact
KB UAV ex.: 6+14 dBi @ 5 Mbps	21840.4 m	21840.4 m	exact
KB UGV ex. (ground mode): 915 MHz, 3+3 dBi	999.6 m	999.6 m	exact
KB UGV ground mode, +4.5 dB gains	1539.3 m	~1593 m	within 3.5%
Terrain: Boulder-Front Range 21 km blocked path	n/a (field-plausible)	36 dB knife-edge, no link	plausible
Custom 1.2 m dish estimate @ 2450 MHz	theory: 27.2 dBi / 7.1 deg	27.2 dBi / 7.1 deg	exact

6. Feature Set

Capability	Description
Interactive planner	Click-to-place draggable nodes; per-node install config; renamable labels; AGL/ASL elevation notation (ground + mast = antenna ASL) live from terrain data
Link analysis	Color-coded links (clear / Fresnel-intruded / blocked / no-close) with distance, best MCS, throughput labels; terrain profile viewer with Fresnel ellipse and worst obstruction
Beam aiming	Beam wedges for directional antennas; drag-to-aim rotation handle; azimuth/tilt fields; pattern losses shown per link
Coverage heatmaps	MCS-colored raster around any node honoring its beam pattern, configurable remote height/gain; sub-second compute
Plan Advisor (new system)	Mission wizard: 6 scenario templates with per-side platform constraints; searches radio x band x bandwidth x antenna space; ranked results; one-click apply
Plan Advisor (extend)	Anchor node + map target: evaluates the real path with the node's actual config; least-change recommendations with full install specs; relay-site search along the profile when needed
Design report export	Self-contained HTML deliverable: objective, map snapshot, equipment/placement table, per-link analysis with embedded profile charts, minimum UAV altitude per link
Save / load	Portable .json layouts + localStorage autosave/restore
Community library	Validated antenna submissions shared to all users (D1-backed API)
Onboarding	First-run spotlight tour (persisted, skippable, replayable); tabbed help center: User Guide, 6 tutorials, 12-entry FAQ

7. Backend, Security & Operations

The Worker serves static assets and two API routes (GET/POST /api/antennas) with "run_worker_first" on /api/* so the SPA fallback cannot shadow them. Submissions are validated server-side: referenceable model numbers (letters + digits, at least 3 distinct characters, placeholder-word blocklist), plausible specs (gain -10..50 dBi, band 30-60000 MHz, beamwidths 1-360 deg), and UNIQUE(manufacturer, model) deduplication. User layouts never leave the browser.

Current posture: the site is public-URL, unauthenticated — appropriate for internal review only. Before customer exposure: add Cloudflare Access (or equivalent), a moderation/approval workflow for community submissions, and rate limiting on the API. OSM public tiles should move to a free-tier commercial provider if usage grows.

8. Known Limitations

- Terrain is ~30 m SRTM: buildings, vegetation and structures are not modeled; urban/forest predictions are optimistic.
- Interference, rain fade and multipath are covered only by the fixed 10 dB fade margin.
- Antenna patterns are parametric (beamwidth-based), not measured 3D pattern files.
- Near-ground model is an empirical fit of the official estimator (~3-4%), not its exact formula.
- Nano2 form factor lacks a published power table; RM-1300/RM-1400 RF tables come from the estimator only.
- Advisor extends from a single anchor node; mesh-wide multi-hop optimization is not yet implemented.

9. Roadmap

Priority	Item
Near-term	Internal validation against field data; Boost BDA as first-class node equipment; community-library moderation queue
Customer-facing gate	Cloudflare Access authentication; branding polish; API rate limits
Mid-term	Mesh-wide advisor (multi-anchor, multi-hop); route sweep for moving platforms (UAV/vessel paths); OSM building clutter
Long-term	Measured 3D antenna pattern import; Longley-Rice (ITM) via WebAssembly; saved cloud projects with sharing

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Repository: github.com/jus24194971/DoodleSim • Live: doodlesim.jus2419497.workers.dev • Data provenance and verification reports: /data in the repository. Predictions are planning-grade; field-verify critical links.